Lab Assignment 09

BRAC UNIVERSITY

Inspiring Excellence

Course Code:	**CSE111**
Course Title:	**Programming Language II**
Topic:	**Inheritance & Method Overriding + Review**
Number of Tasks:	**11**

Task 1

Given the following classes, write the code for the **BBAStudent** class so that the following output is printed when we run the TestStudent class.

Driver Code	Output

`public class Student{`
`  private String name = "Just a Student";`
`  private String department = "nothing";`
 	
`  public void updateDepartment(String dpt){`
`  this.department = dpt;`
`  }`
`  public void updateName(String name){`
`  this.name = name;`
`  }`
`  public void details(){`
`  System.out.println("Name : " + name + "`
`Department: " + department);`
`  }`
`}`
 	
//Tester Class
`public class TestStudent{`
`  public static void main(String [] args){`
`  BBAStudent b1 = new BBAStudent();`
`  BBAStudent b2 = new BBAStudent("Humty Dumty");`
`  BBAStudent b3 = new BBAStudent("Little Bo Peep");`
`  b1.details();`
`  System.out.println("1---------------");`
`  b2.details();`
`  System.out.println("2---------------");`
`  b3.details();`
`  }`
`}`
	`Name : Default Department: BBA`
	`1---------------`
	`Name : Humty Dumty Department: BBA`
	`2---------------`
	`Name : Little Bo Peep Department: BBA`

Task 2

Given the following classes, write the code for the **Vehicle2010** class to print the following output when we run the Vehicle2010User class.

Driver Code	Output

`public class Vehicle{`
`  public int x;`
`  public int y;`
 	
`  public void moveUp(){`
`  y = y+1;`
`  }`
`  public void moveDown(){`
`  y = y-1;`
`  }`
`  public void moveLeft(){`
`  x = x-1;`
`  }`
`  public void moveRight(){`
`  x = x+1;`
`  }`
`  public void position(){`
`  System.out.println("("+ x + ","+ y + ")");`
`  }`
`}`
 	
`//Tester Class`
`public class Vehicle2010User{`
`  public static void main(String[] args){`
`  Vehicle2010 car1 = new Vehicle2010();`
`  car1.position();`
`  car1.moveLowerLeft();`
`  car1.position();;`
 	
`  Vehicle2010 car2 = new Vehicle2010();`
`  car2.position();`
`  car2.moveUpperRight();`
`  car2.position();`
`  car2.moveLowerRight();`
`  car2.position();`
`  }`
`}`
	(0,0)
	(-1,-1)
	(0,0)
	(1,1)
	(2,0)

Task 3

Design the **CheckingAccount** class derived from the Account class with appropriate attributes and properties so that the driver code can generate the output given below.

Driver Code	Output

`public class Account{`
`  public double balance = 0.0;`
 	
`  public Account(double balance){`
`  this.balance = balance;`
`  }`
`  public double showBalance(){`
`  return balance;`
`  }`
`}`
 	
//Tester Class
`public class TestAccount{`
`  public static void main(String [] args){`
`  System.out.println("Total Checking Accounts: "+CheckingAccount.count);`
`  CheckingAccount c1 = new CheckingAccount();`
`  System.out.println("Account Balance: " + c1.showBalance());`
`  CheckingAccount c2 = new CheckingAccount(100.0);`
`  System.out.println("Account Balance: " + c2.showBalance());`
`  CheckingAccount c3 = new CheckingAccount(200.0);`
`  System.out.println("Account Balance: " + c3.showBalance());`
`  System.out.println("Total Checking Accounts: "+CheckingAccount.count);`
`  }`
`}`
	`Total Checking Accounts: 0`
	`Account Balance: 0.0`
	`Account Balance: 100.0`
	`Account Balance: 200.0`
	`Total Checking Accounts: 3`

Task 4

Design the **Dog** and **Cat** class derived from the Animal class with appropriate attributes and properties so that the driver code can generate the output given below.

Driver Code

```
public class Animal {
    public String name;
    public int age;
    public String color;

    public Animal(String name, int age, String color) {
        this.name = name;
        this.age = age;
        this.color = color;
    }
    public void makeSound() {
        System.out.println("Animal makes a sound");
    }
    public String info() {
      return "Name: "+name+"\nAge: "+age+"\nColor: "+color+"\n";
    }
}
//Tester Class
public class AnimalTester {
    public static void main(String[] args) {
      Dog dog = new Dog("Buddy", 5, "Brown", "Bulldog");
      Cat cat = new Cat("Kitty", 3, "White", "Persian");
      System.out.println("1.========");
      System.out.println(dog.info());
      System.out.println("2.========");
      System.out.println(cat.info());
      System.out.println("3.========");
      dog.makeSound();
      System.out.println("4.========");
      cat.makeSound();
    }
}
```

Output

```
1.========
Name: Buddy
Age: 5
Color: Brown
Breed: Bulldog
2.========
Name: Kitty
Age: 3
Color: White
Breed: Persian
3.========
Brown color Buddy is barking
4.========
White color Kitty is meowing
```

Task 5

Implement the design of the **Smartphone** class so that the following output is produced. For simplicity, assume that a smartphone can have a maximum of 10 features.

Driver Code	Output

`public class SmartPhoneTester{`
`  public static void main(String[] args) {`
`  Smartphone s1 = new Smartphone();`
`  System.out.println("1==========");`
`  s1.addFeature("Display", "6.1 inch");`
`  System.out.println("2==========");`
`  s1.updateName("Samsung Note 20");`
`  s1.addFeature("Display", "6.1 inch");`
`  s1.printDetail();`
`  System.out.println("3==========");`
`  Smartphone s2 = new Smartphone("Iphone 12 Pro");`
`  s2.addFeature("Display", "6.2 inch");`
`  s2.addFeature("Ram", "6 GB");`
`  System.out.println("4==========");`
`  s2.printDetail();`
`  s2.addFeature("Display", "Amoled panel");`
`  s2.addFeature("Ram", "DDR5");`
`  System.out.println("5==========");`
`  s2.printDetail();`
`  System.out.println("6==========");`
`  }`
`}`
	`1==========`
	`Feature can not be added without phone name`
	`2==========`
	`Phone Name: Samsung Note 20`
	`Display: 6.1 inch`
	`3==========`
	`4==========`
	`Phone Name: Iphone 12 Pro`
	`Display: 6.2 inch`
	`Ram: 6 GB`
	`5==========`
	`Phone Name: Iphone 12 Pro`
	`Display: 6.2 inch, Amoled panel`
	`Ram: 6 GB, DDR5`
	`6==========`

Task 6

Implement the **Bus** class so that the following output is produced.

Driver Code	Output

`public class BusTester{`
`  public static void main(String args[]){`
`  Bus b1 = new Bus(4, "Jatrabari");`
`  System.out.println("1----------");`
`  Bus b2 = new Bus(10, "Gazipur");`
`  System.out.println("2----------");`
`  b1.addPassenger("Fahim", "Mirpur");`
`  System.out.println("3----------");`
`  b1.addPassenger("Anika", "Jatrabari");`
`  System.out.println("4----------");`
`  b1.addPassenger("Ali");`
`  System.out.println("5----------");`
`  b1.addPassenger("Zafar");`
`  System.out.println("6----------");`
`  b1.addPassenger("Mim", "Badda");`
`  b1.addPassenger("Nowrin");`
`  System.out.println("7----------");`
`  b1.addPassenger("Walid", "Jatrabari");`
`  }`
`}`
	`Capacity: 4`
	`Destination: Jatrabari`
	`1----------`
	`Capacity: 10`
	`Destination: Gazipur`
	`2----------`
	`Sorry Fahim! The bus won't stop at Mirpur`
	`Use another bus.`
	`3----------`
	`Anika is added to the bus.`
	`4----------`
	`Ali is added to the bus.`
	`Ali will get off at the last stop`
	`5----------`
	`Zafar is added to the bus.`
	`Zafar will get off at the last stop`
	`6----------`
	`Sorry Mim! The bus won't stop at Badda`
	`Use another bus.`
	`Nowrin is added to the bus.`
	`Nowrin will get off at the last stop`
	`7----------`
	`Bus is full.`

Task 7

Implement the design of the **Account** class so that the following output is produced:

Driver Code	Output

`public class AccountTester{`
`  public static void main(String[] args) {`
`  System.out.println("Total account holders: " + Account.count);`
`  System.out.println("1===============");`
`  Account p1 = new Account("Abdul",45,"Service Holder",500000);`
`  p1.addMoney(300000);`
`  p1.printDetails();`
`  System.out.println("2===============");`
`  Account p2 = new Account("Rahim",55,"Businessman",700000);`
`  p2.withdrawMoney(700000);`
`  p2.printDetails();`
`  System.out.println("3===============");`
`  Account p3 = new Account("Ashraf",62,"Govt.Officer",200000);`
`  p3.withdrawMoney(250000);`
`  p3.printDetails();`
`  System.out.println("4===============");`
`  System.out.println("Total account holders: " + Account.count);`
`  }`
`}`
	Total account holders: 0
	1===============
	Name: Abdul
	Age: 45
	Occupation: Service Holder
	Total Amount: 800000
	2===============
	Name: Rahim
	Age: 55
	Occupation: Businessman
	Total Amount: 0
	3===============
	Insufficient money for withdrawal!
	Name: Ashraf
	Age: 62
	Occupation: Govt.Officer
	Total Amount: 200000
	4===============
	Total account holders: 3

Task 8

Implement the **Student** class so that the following output is produced.

Driver Code

```
public class StudentTester2{
  public static void main(String[] args) {
    Student s1 = new Student("Naruto", "CSE");
    System.out.println("1----------------");
    s1.individualInfo();
    System.out.println("################");
    Student.totalInfo();
    System.out.println("==================");
    Student s2 = new Student("Sakura", "BBA");
    System.out.println("2----------------");
    s2.individualInfo();
    System.out.println("################");
    Student.totalInfo();
    System.out.println("==================");
    Student s3 = new Student("Shikamaru", "CSE");
    System.out.println("3----------------");
    s3.individualInfo();
    System.out.println("################");
    Student.totalInfo();
    System.out.println("==================");
    Student s4 = new Student("Deidara", "BBA");
    System.out.println("4----------------");
    s4.individualInfo();
    System.out.println("################");
    Student.totalInfo();
  }
}
```

Output

```
Creating Student Number: 1
1----------------
Naruto is from CSE department.
Serial of Naruto among all students' is: 1
Serial of Naruto in CSE department is: 1
################
Total Students: 1
Total CSE Students: 1
Total BBA Students: 0
==================
Creating Student Number: 2
2----------------
Sakura is from BBA department.
Serial of Sakura among all students' is: 2
Serial of Sakura in BBA department is: 1
################
Total Students: 2
Total CSE Students: 1
Total BBA Students: 1
==================
Creating Student Number: 3
3----------------
Shikamaru is from CSE department.
Serial of Shikamaru among all students' is: 3
Serial of Shikamaru in CSE department is: 2
################
Total Students: 3
Total CSE Students: 2
Total BBA Students: 1
==================
Creating Student Number: 4
4----------------
Deidara is from BBA department.
Serial of Deidara among all students' is: 4
Serial of Deidara in BBA department is: 2
################
Total Students: 4
Total CSE Students: 2
Total BBA Students: 2
```

Task 9

```
public class Test1 {
    int x = 2, y = 4, sum = 3;
    int arr[] = {x, y, sum};
    public void methodA(int x) {
        arr[0] += methodB(y, this.x) + methodC(x);
        System.out.println(x + " " + this.x + " " + sum);
        arr[1] += this.x * (++y) / (sum % x);
        System.out.println(y + " " + sum + " " + this.x);
        arr[2] += methodC(x) + methodB(this.x, sum);
        System.out.println(arr[0] + " " + arr[1] + " " + arr[2]);
    }
    public int methodB(int q, int n) {
        int arr2[] = {7, 8};
        int a = (arr2[0]++) - q;
        int b = (++arr2[1]) - n;
        return a + b;
    }
    public int methodC(int z) {
        z = sum + methodB(x, sum) - z;
        return z/2;
    }
}
```

```
public class Tester1{
    public static void main(String [] args){
        Test1 t1 = new Test1();
        t1.methodA(7);
    }
}
```

Outputs		

Task 10

```
1  public class Test3 {
2      int x = 2, y = 4, z = 5;
3      double p = 0.0;
4      public void methodA(int x, int m) {
5          this.x = methodC(this.x);
6          p = x + this.x % m * 3.0;
7          y = y + methodB(x, this.x);
8          System.out.println(this.x +" " + x + y + " " + p) ;
9      }
10     public int methodB(int q, int n) {
11         int arr[] = {3,4,5};
12         arr[0] = arr[0] + this.x + n;
13         arr[1] = q + arr[1];
14         System.out.println(arr[0] +" " + arr[1] + " " + arr[2]) ;
15         return arr[1] + arr[2];
16     }
17     public int methodC(int y) {
18         if(y % 2 == 0) {
19             int temp = methodB(2, y);
20             return temp;
```

```
        }
        else{
            return 4;
       }
    }
}
```

Driver Code	Output		

Test3 t3 = new Test3();
t3.methodA(2,3);
t3.methodB(5,4);			

Task 11

```
public class Quiz3A{
  public static int temp = 4;
  public static int y;
  public int sum;
  public Quiz3A(){
    int y = 7;
    y = temp - 1;
    sum = Quiz3A.temp + 1 + y;
    temp+=2;
  }
```

```
  public Quiz3A(int k){
    temp = temp++;
    sum = ++temp + k;
    Quiz3A.y = (sum++) - 1;
    System.out.println(Quiz3A.y+" "+temp+" "+y);
  }
  public int methodB(int m, int n){
    int x = 0;
    y = this.y + m + (++temp);
    x = x + 2 +  n;
    sum = sum + x + y;
    System.out.println(x + " " + this.y+ " " + sum);
    return sum;
  }
}
```

Driver Code	Output		

public class Tester2{
public static void main(String args[]){
Quiz3A a1 = new Quiz3A();
a1.methodB(1,2);
Quiz3A a2 = new Quiz3A(3);
a2.methodB(2,4);
a1.methodB(2,1);
}
}			